Petrophysics Lecture 5: Gamma Ray & SP Logs

Lithology Identification & Shale Volume (Vsh)

Department / Year: Petroleum Geosciences Department - 2nd Year

Lecture Key Outlines & Core Competencies:

- 1. Principles of Natural Gamma Ray Logging (K, U, Th)
- 2. Calculation of Vshale using Linear and Non-Linear Larionov models
- 3. Spontaneous Potential (SP) Log and Static SP (SSP)

Academic Notice & Verification:

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Section 2: Practical Calculations, Laboratory Equations & Exercises

Core Equations & Petrophysical / Geological Relations:

Equation 1: Phi_total = (V_bulk - V_grain) / V_bulk * 100%

Equation 2: Q = (k * A * delta_P) / (mu * L) [Darcy Law 1D]

Equation 3: Sw = ((a / Phi^m) * (Rw / Rt))^(1 / n) [Archie Saturation]

Equation 4: V_shale = (GR_log - GR_clean) / (GR_shale - GR_clean) [Linear Index]

Laboratory Work Checklist & Submission:

[OK] Step 1: Calibrate electronic calipers and weigh dry clean core plugs.

[OK] Step 2: Record helium injection pressure and stabilization chamber readings.

[OK] Step 3: Plot data on semi-log scale graph and determine trend line slopes.

[OK] Step 4: Upload your lab report directly via PETROGEO Admin & Student portal.